GRAPHICS PROGRAMMING IN LINUX

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This article covers some basics of graphics programming in C.

I was a very avid graphics programmer, using Turbo C (actually, using *graphics.c*, a library usually taught to engineering students during labs). Trying to push *graphics.c* to its limits, I got stuck with its very limited graphics functionality and maximum resolution. When I discussed this with a teacher, he introduced me to OpenGL. I tried a few programs, and got fascinated with the rich graphics experience and platform independence. I thought of sharing my experiences with graphics programming using OpenGL, and contacted the *LFY* editorial team members, who were ready to provide me a platform.

OpenGL basics

OpenGL (Open-source Graphics Library) has three major library header files: *gl.h*, *glu.h* and *glut.h*. The *gl.h* ('graphics library') is a low-level header file, with which you can draw lines, polygons, colour the background or the line, etc. The *glu.h* ('graphics library utility') is a medium-level header. It uses the lower-level OpenGL functions, such as matrices for specific viewing orientation, rendering surfaces, etc. The *glut.h* ('graphics library utility tool-kit') is a high-level library file and a window system-independent tool-kit to hide the complexity of different window system APIs.

Installation

Now, we need a Linux-based OS (I use Kubuntu 11.10); a compiler (GCC); an editor (Kate, Gedit, Kwrite, etc); the OpenGL header files... and some basic knowledge of C programming.

Make sure your PC has an Internet connection. Open a terminal and run *'sudo apt-get install gcc'*, entering the password when prompted, to install GCC. Next, for the editor, run *'sudo apt-get install kate'* (or *gedit*, *kwrite* or whatever